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Master Thesis

CLASSIFICATION OF HOMEOMORPHISMS ON THREE–DIMENSIONAL MANIFOLDS WITH PSEUDO–ANOSOV ATTRACTORS AND REPELLERS

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Abstract

The thesis is devoted to the study of orientation-preserving homeomorphisms of three-dimensional manifolds with a nonwandering set consisting of a finite number of two-dimensional attractors and repellers, each of which is a disjoint union of cylindrically embedded closed surfaces, the restriction of some power to each of which is topologically conjugate to an orientation-preserving pseudo-Anosov homeomorphism. Topological classification of model homeomorphisms realized on each manifold admitting homeomorphisms of the class under study is obtained. It is proved that a homeomorphism of the class under consideration is topologically conjugate to a model mapping if and only if it has a one-dimensional invariant foliation.

1 Introduction

1.1 History of the issue

In works by V.Z. Grines, Yu. A. Levchenko, V. S. Medvedev and O. V. Pochinka [1], [2], [3] the dynamics of A-diffeomorphisms of three-dimensional manifolds with a two-dimensional surface nonwandering set was studied. For such mappings the topology of the supporting manifold was investigated, model mappings were constructed, the ambient Ω-conjugacy of an arbitrary diffeomorphism of the class under consideration with the model mapping and the topological conjugacy of a structurally stable diffeomorphism of the class under consideration with the model mapping were proved. According to V.Z. Grines, V.S. Medvedev, E.V. Zhuzhoma [4], the number of connected components of the nonwandering set of mappings of the studied class is finite and each of them is a cylindrically embedded two-dimensional torus ([4, Theorem 1]) such that the restriction of some degree of the mapping to it is topologically conjugate to an Anosov diffeomorphism (см. [4, Theorem 2]). It follows from the work by R.V. Plykin [5] that with such a structure of the nonwandering set, each of its connected components is either an attractor or a repeller (см. [5, Theorem 3]).

This thesis is a generalization of the results of works [1], [2], [3] to the class of mappings $\mathcal{G}$, which is introduced below.

1.2 Definition of the considered class of mappings

Denote by S_g a closed orientable surface of genus g.

A homeomorphism $P\colon S_g \to S_g$ is called a *generalized pseudo-Anosov homeomorphism* with *dilatation* $\lambda > 1$ if on the surface S_g there exists a pair of P-invariant transverse foliations $\mathcal{F}^s_P$, $\mathcal{F}^u_P$ with a finite set of saddle singularities S and transverse measures μ_s, μ_u such that $\mu_s(P(\alpha)) = \lambda\mu_s(\alpha)$ $(\mu_u(P(\alpha)) = \lambda^{-1}\mu_u(\alpha))$ for any arc α transverse to $\mathcal{F}^s_P$ $(\mathcal{F}^u_P)$. If the foliations $\mathcal{F}^s_P$, $\mathcal{F}^u_P$ do not have saddle singularities with one separatrix, then the homeomorphism $P\colon S_g \to S_g$ is called *pseudo-Anosov*. It follows from the Euler-Poincare formula (see Propositions 2.1 and 2.2) that pseudo-Anosov homeomorphisms exist only on surfaces of genus greater than one.

A subspace X of a topological space Y is called a *cylindrical embedding in Y of a topological space $\bar{X}$* if there exists a homeomorphism onto the image $h : \bar{X} \times [-1, 1] \to Y$ such that $X = h(\bar{X} \times \{0\})$.

An invariant set B of a homeomorphism f is called an *attractor* if there exists a closed neighborhood U of B such that $f(U) \subset int\, U$, $\bigcap\limits_{j \geq 0} f^j(U) = B$. An attractor for a homeomorphism f^{-1} is called a *repeller* of the homeomorphism f.

Let us denote by $\mathcal{G}$ the set of orientation-preserving homeomorphisms f of a closed orientable topological 3-manifold M^3, the nonwandering set $NW(f)$ of which consists of a finite number of connected components $B_0, \ldots, B_{m-1}$ satisfying for any $i \in \{0, \ldots, m-1\}$ the following conditions:

1. B_i is a cylindrical embedding of a closed orientable surface of genus greater than one;

2. there exists a natural number k_i such that $f^{k_i}(B_i) = B_i$, $f^k(B_i) \neq B_i$ for any $k \leq k_i$ and the restriction of the mapping $f^{k_i}|_{\mathcal{B}_i}$ is topologically conjugate to an orientation-preserving pseudo-Anosov homeomorphism;

3. B_i is either an attractor or a repeller of the homeomorphism f^{k_i}.

A number of results related to the dynamics of mappings of class $\mathcal{G}$ have been obtained previously (Propositions 1.1-1.5).

1.3 Main results

The main results of the thesis are Theorems 1-2. To begin with, let us introduce the necessary notation.

Let $J\colon S_g \to S_g$ be a homeomorphism. We define the homeomorphism

$$\gamma\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$$

by the formula

$$\gamma(z, r) = (J(z), r - 1).$$

Denote by M_J the quotient space of $S_g \times \mathbb{R}$ by the action of the group

$$\Gamma = \{\gamma^i, i \in \mathbb{Z}\}$$

with the natural projection

$$p_J \colon S_g \times \mathbb{R} \to M_J.$$

Proposition 1.1. *[6, Theorem 1.1] A topological manifold admits a homeomorphism from the class $\mathcal{G}$ if and only if it is homeomorphic to M_J, where $J \colon S_g \to S_g$ is an orientation-preserving homeomorphism*[1]*, commuting with a pseudo-Anosov homeomorphism.*

A homeomorphism $f \colon X \to X$ is called *periodic* if there exists $m \in \mathbb{N}$ such that $f^m = id_X$.

Proposition 1.2. *[7, Theorem 1] A homeomorphism commuting with a pseudo-Anosov map is either pseudo-Anosov or periodic homeomorphism.*

The simplest representatives of the class $\mathcal{G}$ are homeomorphisms of the set Φ, which are constructed as follows.

Let us represent the circle as a subset of the complex plane

$$\mathbb{S}^1 = \{e^{i2\pi\theta} | 0 \leq \theta < 1\}.$$

We define the covering

$$p \colon \mathbb{R} \to \mathbb{S}^1$$

by the formula $p(r) = e^{i2\pi r}$.

A set of numbers (n, k, l) is called *correct* if the following conditions hold:

1. $n, k \in \mathbb{N}$;
2. if $k = 1$, then $l = 0$;
3. if $k > 1$, then $l \in \{1, \ldots, k-1\}$ and is coprime to k.

For each correct set (n, k, l) we define a diffeomorphism $\bar{\varphi}_{n,k,l} \colon \mathbb{R} \to \mathbb{R}$ by the formula

$$\bar{\varphi}_{n,k,l}(r) = r + \frac{1}{4\pi nk} sin(2\pi nkr) + \frac{l}{k}.$$

[1]The statement of Theorem 1.1 in [6] omits the fact that J preserves the orientation of S_g, but it is necessary since homeomorphisms of class $\mathcal{G}$ are defined on orientable manifolds.

Since $\bar{\varphi}_{n,k,l}(r)+1=\bar{\varphi}_{n,k,l}(r+1)$, the diffeomorphism $\bar{\varphi}_{n,k,l}$ is a lift of the circle diffeomorphism $\varphi_{n,k,l}(s)=p(\bar{\varphi}_{n,k,l}(p^{-1}(s)))$, where $p^{-1}(s)$ is the full preimage of the point $s\in\mathbb{S}^1$ [8, Proposition 10.2.26].

Consider a pseudo-Anosov homeomorphism $P\colon S_g\to S_g$, an orientation-preserving homeomorphism $J\colon S_g\to S_g$ that commutes with it, and a correct set of numbers n,k,l such that the homeomorphism J^lP^k is an orientation-preserving pseudo-Anosov homeomorphism.

Let us define the mapping $\bar{\varphi}_{P,J,n,k,l}\colon S_g\times\mathbb{R}\to S_g\times\mathbb{R}$ by the formula

$$\bar{\varphi}_{P,J,n,k,l}(z,r)=(P(z),\bar{\varphi}_{n,k,l}(r)).$$

It is directly verified that $\bar{\varphi}_{P,J,n,k,l}\gamma=\gamma\bar{\varphi}_{P,J,n,k,l}$. Then, according to [8, Proposition 10.2.26], the orientation-preserving homeomorphism $\varphi_{P,J,n,k,l}:M_J\to M_J$ is well defined and given by the formula

$$\varphi_{P,J,n,k,l}(w)=p_J(\bar{\varphi}_{P,J,n,k,l}(p_J^{-1}(w))),$$

where $w\in M_J$ and $p_J^{-1}(w)$ is the full preimage of the point $w\in M_J$. Homeomorphisms of the form $\varphi_{P,J,n,k,l}$ will be called *model*.

Let Φ denote the set of all model homeomorphisms.

Proposition 1.3. *[7, Theorem 2]* $\Phi\subset\mathcal{G}$.

For $\varphi_{P,J,n,k,l}\in\Phi$ and $i\in\{0,1,\dots,2nk-1\}$, we set

$$\mathcal{B}_i=p_J\Big(S_g\times\Big\{\frac{i}{2nk}\Big\}\Big).$$

It follows from the proof of [7, Theorem 2] that the following statement is true:

Proposition 1.4. *Let* $\varphi=\varphi_{P,J,n,k,l}\in\Phi$. *Then the following statements hold:*

1. *the nonwandering set* $NW(\varphi)$ *of the homeomorphism* φ *consists of* $2nk$ *connected components* $\mathcal{B}_0,\mathcal{B}_1,\dots,\mathcal{B}_{2nk-1}$ *of period* k *(i.e.* $\varphi^k(\mathcal{B}_i)=\mathcal{B}_i$ *and* $\varphi^{k'}(\mathcal{B}_i)\neq\mathcal{B}_i$ *for any* $i\in\{0,\dots,2nk-1\}$ *and a natural* $k'<k$*);*

2. *for any* $i \in \{0, 1, \dots, 2nk-1\}$ *the restriction of* $\varphi|_{\mathcal{B}_i}$ *to the connected component* $\mathcal{B}_i$ *is topologically conjugate to the orientation-preserving pseudo-Anosov homeomorphism* $J^l P^k$.

Homeomorphisms $f\colon X \to X$ and $f'\colon X' \to X'$ are called *topologically conjugate* if there exists a homeomorphism $h\colon X \to X'$ such that $hf = f'h$.

Theorem 1. [2] *Homeomorphisms* $\varphi_{P,J,n,k,l}$ *and* $\varphi_{P',J',n',k',l'}$ *are topologically conjugate if and only if* $n = n'$, $k = k'$ *and there exists a homeomorphism* $H\colon S_g \to S_g$ *such that one of the following conditions holds:*

1. $l = l' \geq 0$, $HP = P'H$ *and* $HJ = J'H$;
2. $l = l' = 0$, $HP = P'H$ *and* $HJ = J'^{-1}H$;
3. $l + l' = k$, $HP = J'P'H$ *and* $HJ = J'^{-1}H$.

Homeomorphisms $f\colon X \to X$ and $f'\colon X' \to X'$ are called ambient Ω-conjugate if there exists a homeomorphism $h\colon X \to X'$ such that $h(NW(f)) = NW(f')$ and $hf|_{NW(f)} = f'h|_{NW(f)}$.

Proposition 1.5. *Let* $f\colon M^3 \to M^3$ *be a homeomorphism from the class* $\mathcal{G}$. *Then the following statements hold:*

1. *[6, Lemma 2.1] the boundary of each connected component of the set* $M^3 \setminus (NW(f))$ *consists of exactly one connected component of the attractor* f *and one connected component of the repeller* f;
2. *[7, Theorem 3]* f *is ambient* Ω*-conjugate to a homeomorphism from the class* Φ.

It follows from Propositions 1.4 and 1.5 that the nonwandering set $NW(f)$ of the homeomorphism $f \in \mathcal{G}$ consists of $2nk$ connected components of period k ($n, k \in \mathbb{N}$) and the boundary of each connected component of the set $M^3 \setminus NW(f)$ consists of exactly two connected components. Denote by $\mathcal{G}_* \subset \mathcal{G}$ the

[2] In the work [6], in the classification of model mappings in point 3, an inaccuracy was made. Theorem 1 is a corrected version of it.

subset of homeomorphisms f such that in each connected component V of the set $M^3 \setminus NW(f)$ there exists a one-dimensional f^k-invariant foliation, each leaf of which is an open arc having exactly two boundary points lying on two different connected components of the set ∂V. The union of the closures of all such arcs on all connected components of $M^3 \setminus NW(f)$ forms an f-invariant one-dimensional foliation on M^3, which we denote by $\mathcal{L}_f$.

Note that, by construction, $\Phi \subset \mathcal{G}_*$.

Theorem 2. *A homeomorphism $f \in \mathcal{G}$ is topologically conjugate to a homeomorphism from the class Φ if and only if $f \in \mathcal{G}_*$.*

The results known for pseudo-Anosov homeomorphisms made it possible to prove that the dynamics of mappings of the class $\mathcal{G}$ on a nonwandering set, up to topological conjugacy, has the same structure as the dynamics of model mappings of the set Φ. If an invariant one-dimensional foliation exists, then the dynamics of mappings of the class $\mathcal{G}$ coincides with the dynamics of model mappings on the entire manifold. In particular, it follows from Propositions 1.4 and 1.5 that the restrictions of a mapping from class $\mathcal{G}$ to different connected components of its nonwandering set are topologically conjugate homeomorphisms (for the same power, equal to the period of the connected components of the nonwandering set of the homeomorphism). If we assume conjugacy with generalized pseudo-Anosov homeomorphisms on the connected components of a nonwandering set of homeomorphisms of the class under consideration, then we can obtain pairwise topologically non-conjugate mappings on each connected component. In section 5 such example is constructed.

2 Auxiliary Statements

2.1 Generalized pseudo-Anosov homeomorphisms

This section contains propositions related to generalized pseudo-Anosov homeomorphisms, which are necessary for proving the main theorems (for more details on generalized pseudo-Anosov homeomorphisms, see [9] and [10]). Let $P\colon S_g \to S_g$ be a generalized pseudo-Anosov homeomorphism. For $q \in \mathbb{N}$, let d_q denote the number of saddle singularities of invariant foliations of a pseudo-Anosov homeomorphism P that have exactly q separatrices. The sequence $\mathcal{D}_P = \{d_q : q \in \mathbb{N}\}$ is called the *singular type* of P. Since the sequence $\mathcal{D}_P$ contains only a finite number of nonzero elements, for brevity we will indicate only the nonzero elements of this sequence below.

Proposition 2.1. *Euler-Poincare formula [9, Theorem 5.4] Let $\mathcal{D}_P = \{d_q : q \in \mathbb{N}\}$ be the singular type of the generalized pseudo-Anosov homeomorphism $P\colon S_g \to S_g$. Then*

$$d_1 - \sum_{q\geq 3}(q-2)d_q = 2(2-2g).$$

Proposition 2.2. *[11] For an orientable surface, there exists a generalized pseudo-Anosov homeomorphism of any singular type admitted by the Euler-Poincare formula, with the exception of the singular types $\{d_1 = d_3 = 1\}$ and $\{d_3 = d_5 = 1\}$ in the case of surfaces of genus 1 and 2, respectively.*

Proposition 2.3. *[10, Theorem 12.5] Two homotopic pseudo-Anosov homeomorphisms are topologically conjugate by a homeomorphism isotopic to the identity.*

Proposition 2.4. *[12, Corollary] Let $P_1, P_2\colon S_g \to S_g$ be pseudo-Anosov homeomorphisms and there exist homeomorphisms $H_1, H_2\colon S_g \to S_g$ such that $H_1P_1 = P_2H_1$, $H_2P_1 = P_2H_2$. Then if the homeomorphism H_1 is homotopic to the homeomorphism H_2, then $H_1 = H_2$.*

2.2 Group action on a topological space

Let us recall some facts related to the group action on a topological space (for more details, see [8]).

A path $c \subset X$ is a continuous image of a segment $[0,1]$ in X. Denote by $c(t)$ the image of $t \in [0,1]$ in X. A path c is called a loop if $c(0) = c(1)$. Denote by $[c] \in \pi_1(X,x)$ the class of loops homotopic to the loop $c \subset X$ with the beginning and end at the point $x \in X$.

For a continuous mapping $f\colon X \to Y$ of a topological space X to a topological space Y, we set $f^{-1}(V) = \{x \in X | h(x) \in V\}$ (the full preimage of the set $V \subset Y$) and denote by f_* the fundamental group homomorphism $f_*\colon \pi_1(X,x) \to \pi_1(Y, f(x))$ induced by the mapping f at $x \in X$.

Let the group Γ act freely and discontinuously on a Hausdorff space X such that the orbit space X/Γ is connected. From the definition of the projection $p_{X/\Gamma}\colon X \to X/\Gamma$ it follows that $p^{-1}_{X/\Gamma}(x)$ is the orbit of some point $\bar{x} \in p^{-1}_{X/\Gamma}(x)$. Let c be some loop in X/Γ such that $c(0) = c(1) = x$. By the monodromy theorem, there exists a unique path $\bar{c}$ in X starting at $\bar{x}$ ($\bar{c}(0) = \bar{x}$), which is a lift of c. Therefore, there exists an element $\tilde{\gamma} \in \Gamma$ such that $\bar{c}(1) = \tilde{\gamma}(\bar{x})$ and the mapping $\eta_{X/\Gamma,\bar{x}}\colon \pi_1(X/\Gamma, x) \to \Gamma$ is well defined by the formula $\eta_{X/\Gamma,\bar{x}}([c]) = \tilde{\gamma}$.

The mapping $\eta_{X/\Gamma,\bar{x}}\colon \pi_1(X/\Gamma, x) \to \Gamma$ is a nontrivial homomorphism ([8, Proposition 10.2.23]), called the *homomorphism induced by the covering* $p_{X/\Gamma}\colon X \to X/\Gamma$.

Let Γ be an abelian group and $\bar{c}'$ be the lift of the path $c \in \pi_1(X/\Gamma, x)$ starting at $\bar{x}' = \bar{c}'(0)$ ($\bar{x}' \neq \bar{x}$) and $\tilde{\gamma}'(\bar{x}') = \bar{c}'(1)$. Since there is a unique element $\tilde{\gamma}'' \in \Gamma$ such that $\tilde{\gamma}''(\bar{x}) = \bar{x}'$, by the monodromy theorem $\tilde{\gamma}''(\bar{c}) = \bar{c}'$. Then $\tilde{\gamma}''\tilde{\gamma} = \tilde{\gamma}'\tilde{\gamma}''$ and, consequently, $\tilde{\gamma}' = \tilde{\gamma}$. Thus, $\eta_{X/\Gamma,\bar{x}} = \eta_{X/\Gamma,\bar{x}'}$. Therefore, we will omit the index $\bar{x}$ in the notation of the epimorphism $\eta_{X/\Gamma,\bar{x}}$ in the case of an Abelian group Γ and write $\eta_{X/\Gamma}$.

Proposition 2.5. *[8, Proposition 10.2.26] Let Γ and Γ' be cyclic groups acting freely and discontinuously on a Γ, Γ'-space X, γ be a generator of Γ, and let there exist an isomorphism $\tau\colon \Gamma \to \Gamma'$. Then the following statements are true:*

1. *If $\bar{h}\colon X \to X$ is a homeomorphism such that $\bar{h}(\gamma(\bar{x})) = \tau(\gamma(\bar{h}(\bar{x})))$ for any $\bar{x} \in X$, then the mapping $h\colon X/\Gamma \to X/\Gamma'$ given by the formula $h = p_{X/\Gamma'}(\bar{h}(p^{-1}_{X/\Gamma}(x)))$ is a homeomorphism and $\tau\eta_{X/\Gamma} = \eta_{X/\Gamma'}h_*$;*

2. *If $h\colon X/\Gamma \to X/\Gamma'$ is a homeomorphism such that $\tau\eta_{X/\Gamma} = \eta_{X/\Gamma'}h_*$, then there exists a unique homeomorphism $\bar{h}\colon X \to X$ that is a lift of h such*

that $\bar{h}(\gamma(\bar{x})) = \tau(\gamma(\bar{h}(\bar{x})))$, $\bar{h}(\bar{x}) = \bar{x}'$ *for* $\bar{x} \in X$ *and* $\bar{x}' \in p^{-1}_{X/\Gamma'}(x')$, *where* $x' = h(p_{X/\Gamma}(\bar{x}))$.

A group G is called a *semidirect product of subgroups* $H \subset G$ *and* $N \subset G$ if N is a normal subgroup and any element $g \in G$ is uniquely represented as $g = hn$, where $h \in H$, $n \in N$. We will write $G = H \ltimes N$.

Let $J\colon S_g \to S_g$ be a homeomorphism. Recall that $\mathcal{B}_0 = p_J(S_g \times \{0\})$ and the manifold $M_J = S_g \times \mathbb{R}/\Gamma$ is the quotient space of the manifold $S_g \times \mathbb{R}$ by the action of the cyclic group of homeomorphisms Γ with the generator $\gamma\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ given by the formula $\gamma(z, r) = (J(z), r - 1)$.

The epimorphism induced by the covering $p_J\colon S_g \times \mathbb{R} \to M_J$ will be denoted by

$$\eta_J\colon M_J \to \Gamma.$$

Denote by ι_J the inclusion map

$$\iota_J\colon \mathcal{B}_0 \to M_J.$$

Proposition 2.6. *Let* $J\colon S_g \to S_g$ *be a homeomorphism. Then the following statements hold:*

1. *[13, Theorem 17.7] there exists an element* $[a] \in \pi_1(M_J)$ *of infinite order such that* $\pi_1(M_J) = \langle [a] \rangle \ltimes \iota_{J*}(\pi_1(\mathcal{B}_0))$*;*

2. *if* $[c] = [a]^k[b] \in \pi_1(M_J)$, *where* $k \in \mathbb{Z}$, $[b] \in \iota_{J*}(\pi_1(\mathcal{B}_0))$, *then*

$$\eta_J([c]) = \gamma^{\pm k}.$$

2.3 Foliations and bundles

Let us recall the definitions related to foliations and bundles (for more details, see [14]).

A family $\mathcal{F} = \{L_i; i \in \Upsilon\}$ of path-connected subsets of a topological manifold X of dimension n is called a *k-dimensional foliation* if it satisfies the following three conditions:

- $L_i \cap L_j = \emptyset$ for any $i, j \in \Upsilon$ such that $i \neq j$;
- $\bigcup\limits_{i \in \Upsilon} L_i = X$;
- for any point $x \in X$ we can choose a local map (U_x, φ_x), $x \in U_x$, such that if $U_x \cap L_i \neq \emptyset$, $i \in \Upsilon$, then the path-connected components of the set $\varphi_x(U_x \cap L_i)$ have the form $\{(x_1, x_2, \ldots, x_n) \in \varphi_x(U_x);\ x_{k+1} = c_{k+1}, x_{k+2} = c_{k+2}, \ldots, x_n = c_n\}$, where $c_{k+1}, c_{k+2}, \ldots, c_n$ are constants.

An element of the foliation $\mathcal{F}$ is called a *leaf*.

A foliation $\mathcal{F}$ is called invariant under a homeomorphism $f\colon X \to X$ if f maps a leaf of $\mathcal{F}$ to a leaf of $\mathcal{F}$.

Let X be a topological manifold of dimension n. Then a k-dimensional foliation $\mathcal{F}_1$ is called *transversal* to a $(n-k)$-dimensional foliation $\mathcal{F}_2$ if for any point $x \in X$ there exists a local chart (U_x, φ_x) such that $\varphi_x(L_x^1 \cap U_x) = \{(x_1, x_2, \ldots, x_n) \in \varphi_x(U_x);\ x_{k+1} = x_{k+2} = \cdots = x_n = 0\}$ and $\varphi_x(L_x^2 \cap U_x) = \{(x_1, x_2, \ldots, x_n) \in \varphi_x(U_x);\ x_1 = x_2 = \cdots = x_k = 0\}$, where L_x^1 (L_x^2) is the leaf of the foliation $\mathcal{F}_1$ ($\mathcal{F}_2$) containing the point x.

A bundle of a topological manifold X over a space B with projection ξ is a triple of the form (X, ξ, B), where $\xi\colon X \to B$ is a continuous mapping. The preimage $\xi^{-1}(b)$ is called *a leaf of the bundle* (X, ξ, B) *over the point* $b \in B$.

An isomorphism of bundles (X, ξ, B) and (X', ξ', B') is a pair of homeomorphisms $h\colon X \to X'$, $H\colon B \to B'$ such that $\xi' h = H\xi$. The homeomorphism h maps the leaf of (X, ξ, B) to the leaf of (X', ξ', B').

A triple of the form $(B \times F, p_B, B)$, where $p_B\colon B \times F \to B$ is the canonical projection, is called *the standard trivial bundle*.

3 Model Mappings

Recall that the homeomorphism $\varphi = \varphi_{P,J,n,k,l} \in \Phi$ is given by the formula

$$\varphi(w) = p_J(\bar{\varphi}(p_J^{-1}(w))),$$

where

$$\bar{\varphi}(z,r) = (P(z), \bar{\varphi}_{n,k,l}(r)), \tag{1}$$

and is defined on the manifold $M_J = S_g \times \mathbb{R}/\Gamma$ with the natural projection

$$p_J \colon S_g \times \mathbb{R} \to M_J,$$

where

$$\Gamma = \{\gamma^i | i \in \mathbb{Z}\},$$

$$\gamma(z,r) = (J(z), r-1). \tag{2}$$

The covering p_J induces a homomorphism $\eta_J \colon \pi_1(M_J) \to \Gamma$.

For $i \in \{0, \dots, 2nk-1\}$ we assume that $\mathcal{B}_i = p_J(S_g \times \{\frac{i}{2nk}\})$ and $\iota_J \colon \mathcal{B}_0 \to M_J$ is an inclusion map.

Everywhere below it is assumed that for a homeomorphism $\varphi' = \varphi_{P',J',n',k',l'} \in \Phi$ all objects associated with it have similar notations with a prime.

Lemma 3.1. *Let homeomorphisms φ and φ' be topologically conjugate. Then the following statements are true:*

1. *$g = g'$, $n = n'$, $k = k'$;*
2. *there exist homeomorphisms $h \colon M_J \to M_{J'}$ and its lift $\bar{h} \colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ such that $h\varphi = \varphi' h$, $\bar{h}(S_g \times \{0\}) = S_g \times \{0\}$ and $\bar{h}\gamma = \gamma'^{\pm 1}\bar{h}$.*

Proof. Let homeomorphisms φ and φ' be topologically conjugate by a homeomorphism $h' \colon M_J \to M_{J'}$, i.e.

$$h'\varphi = \varphi' h'. \tag{3}$$

The nonwandering set $NW(\varphi)$ ($NW(\varphi')$) of the homeomorphism φ (φ') consists of $2nk$ ($2n'k'$) connected components of period k (k') homeomorphic to S_g ($S_{g'}$)

(see Proposition 1.4). Since the conjugating homeomorphism maps orbits to orbits and a nonwandering set to a nonwandering set, $g = g'$, $n = n'$, and $k = k'$. Item 1 is proved.

We prove item 2. Recall that $NW(\varphi) = \mathcal{B}_0 \sqcup \cdots \sqcup \mathcal{B}_{2nk-1}$ and $NW(\varphi') = \mathcal{B}'_0 \sqcup \cdots \sqcup \mathcal{B}'_{2nk-1}$ (see Proposition 1.4).

The homeomorphism h' maps the connected components of the nonwandering set $NW(\varphi)$ of the homeomorphism φ to the connected components of the nonwandering set $NW(\varphi')$ of the homeomorphism φ'. Let $h'(\mathcal{B}_0) = \mathcal{B}'_m$.

We define a homeomorphism $\bar{\alpha}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ by the formula

$$\bar{\alpha}(z, r) = \Big(z, r - \frac{m}{2nk}\Big). \tag{4}$$

It is directly verified that $\bar{\alpha}\gamma' = \gamma'\bar{\alpha}$, $\bar{\alpha}\bar{\varphi}' = \bar{\varphi}'\bar{\alpha}$. Consequently, the mapping $\bar{\alpha}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ projects into a homeomorphism $\alpha\colon M_{J'} \to M_{J'}$, commuting with φ' by the formula $\alpha(w) = p_{J'}(\bar{\alpha}(p_{J'}^{-1}(w)))$ (see Proposition 2.5). Thus, $\bar{\alpha}(S_g \times \{\frac{m}{2nk}\}) \overset{(4)}{=} S_g \times \{0\}$ and $\alpha(\mathcal{B}'_m) = \mathcal{B}'_0$. We put

$$h = \alpha h' \tag{5}$$

and obtain that $h(\mathcal{B}_0) = \mathcal{B}'_0$, $h\varphi h^{-1} = \varphi'$.

Consider the point $w \in \mathcal{B}_0$. It follows from Proposition 2.6 that there exists an element $[a] \in \pi_1(M_J, w)$ such that any element $[c] \in \pi_1(M_J, w)$ is uniquely represented as

$$[c] = [a]^k[b],$$

where $k \in \mathbb{Z}$, $[b] \in \iota_{J*}(\pi_1(\mathcal{B}_0, w))$, and

$$\eta_J([c]) = \gamma^k. \tag{6}$$

Let us put $w' = h(w)$, $\bar{w} = p_J^{-1}(w) \cap (S_g \times \{0\})$ and $\bar{w}' = p_{J'}^{-1}(w') \cap (S_g \times \{0\})$. Homeomorphisms $h\colon M_J \to M_{J'}$, $h|_{\mathcal{B}_0}\colon \mathcal{B}_0 \to \mathcal{B}'_0$ induce isomorphisms $h_*\colon \pi_1(M_J, w) \to \pi_1(M_{J'}, w')$, $h_*|_{\iota_{J*}(\pi_1(\mathcal{B}_0, w))}\colon \iota_{J*}(\pi_1(\mathcal{B}_0, w)) \to \iota_{J'*}(\pi_1(\mathcal{B}'_0, w'))$. Then it is directly verified that any element of $[c'] \in \pi_1(M_{J'})$ is uniquely represented as $[c'] = [a']^k[b']$, where $[a']^k \in h_*(\langle[a]\rangle)$, $[b'] \in h_*(\iota_{J'*}(\pi_1(\mathcal{B}'_0, w')))$. There-

fore, $\pi_1(M_{J'}) = \langle [a'] \rangle \ltimes \iota_{J'*}(\pi_1(\mathcal{B}'_0))$. By Proposition 2.6,

$$\eta_J([c']) = \gamma'^{\pm k}. \tag{7}$$

Let τ be an isomorphism $\tau\colon \Gamma \to \Gamma'$ such that

$$\tau(\gamma) = \gamma'^{\pm 1}. \tag{8}$$

Then $\tau(\eta_J([c])) \overset{(6)}{=} \tau(\gamma^k) \overset{(8)}{=} \gamma'^{\pm k}$ and $\eta_{J'}(h_*([c])) = \eta_{J'}([c']) \overset{(7)}{=} \gamma'^{\pm k}$. Therefore, $\tau\eta_J = \eta_{J'}h_*$ and, by Proposition 2.5, there exists a unique homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ that is a lift of h such that $\bar{h}(\bar{w}) = \bar{w}'$ and $\bar{h}\gamma = \gamma'^{\pm 1}\bar{h}$. Since $h(\mathcal{B}_0) = \mathcal{B}'_0$ and $\bar{w}, \bar{w}' \in S_g \times \{0\}$, we get that $\bar{h}(S_g \times \{0\}) = S_g \times \{0\}$.

□

Everywhere below, if homeomorphisms φ and φ' are topologically conjugate by assumption, then we will assume that homeomorphisms J and J' are defined on the same surface S_g.

We define the set

$$\mathcal{I} = \left\{ \frac{i}{2nk} \middle| i \in \mathbb{Z} \right\}.$$

Lemma 3.2. *Let homeomorphisms φ and φ' be topologically conjugate. Then either $l = l'$ or $k = l + l'$. Moreover, if $h\colon M_J \to M_{J'}$ is a homeomorphism such that $h\varphi = \varphi' h$ and it has a lift $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ that has the property $\bar{h}(S_g \times \{0\}) = S_g \times \{0\}$, then one of the following statements is true:*

1. *$l = l' \geq 0$, $\bar{h}\gamma = \gamma'\bar{h}$, $\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1}$ and $\bar{h}(S_g \times \{r\}) = S_g \times \{r\}$ $\forall r \in \mathcal{I}$;*
2. *$l = l' = 0$, $\bar{h}\gamma = \gamma'^{-1}\bar{h}$, $\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1}$ and $\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}$ $\forall r \in \mathcal{I}$;*
3. *$l + l' = k$, $\bar{h}\gamma = \gamma'^{-1}\bar{h}$, $\gamma'\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1}$ and $\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}$ $\forall r \in \mathcal{I}$.*

Proof. Let homeomorphisms φ and φ' be topologically conjugate. Then, by item 2 of Lemma 3.1, there exists the homeomorphism $h\colon M_J \to M_{J'}$ and its lift $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ such that $h\varphi = \varphi' h$ and $\bar{h}(S_g \times \{0\}) = S_g \times \{0\}$.

Consider the lift $\bar{\varphi}$ of the homeomorphism φ. Then the map

$$\bar{\varphi}'_0 = \bar{h}\bar{\varphi}\bar{h}^{-1} \tag{9}$$

is a lift of φ'. Since $\bar{\varphi}'_0$ is a lift of the homeomorphism φ' onto $S_g \times \mathbb{R}$ with respect to the covering $p_{j'}$, it follows that

$$\bar{\varphi}'_0 = \gamma'^j \bar{\varphi}', \tag{10}$$

where $j \in \mathbb{Z}$. Then

$$\bar{\varphi}'_0(S_g \times \{0\}) = \gamma'^j(\bar{\varphi}'(S_g \times \{0\})) \overset{(1),(2)}{=} S_g \times \left\{\frac{l'}{k} - j\right\}. \tag{11}$$

Since the conjugating homeomorphism h maps the connected components of the nonwandering set $NW(\varphi)$ to the connected components of the nonwandering set $NW(\varphi')$ and the wandering set to the wandering set, then for any $r \in \mathcal{I}$ one of cases is possible: a) $\bar{h}(S_g \times \{r\}) = S_g \times \{r\}$, b) $\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}$.

Let us consider case a):

$$\bar{h}(S_g \times \{r\}) = S_g \times \{r\}. \tag{12}$$

Note that $\bar{\varphi}'_0(S_g \times \{0\}) \overset{(9)}{=} \bar{h}(\bar{\varphi}(\bar{h}^{-1}(S_g \times \{0\}))) \overset{(12)}{=} \bar{h}(\bar{\varphi}(S_g \times \{0\})) \overset{(1)}{=} \bar{h}(S_g \times \{\frac{l}{k}\}) \overset{(12)}{=} S_g \times \{\frac{l}{k}\}$. Taking into account the formula (11), we obtain that $S_g \times \{\frac{l}{k}\} = S_g \times \{\frac{l'}{k} - j\}$. Since $\frac{l}{k}, \frac{l'}{k} \in [0, 1)$, the equality $\frac{l'}{k} - j = \frac{l}{k}$ is satisfied only for $j = 0$, that is, for

$$\bar{\varphi}'_0 = \bar{\varphi}'. \tag{13}$$

Consequently, $l = l'$.

Since $\bar{h}(\gamma(\bar{h}^{-1}(S_g \times \{0\}))) \overset{(12)}{=} \bar{h}(\gamma(S_g \times \{0\})) \overset{(2)}{=} \bar{h}(S_g \times \{-1\}) \overset{(12)}{=} S_g \times \{-1\} \overset{(2)}{=} \gamma'(S_g \times \{0\})$ and, according to Lemma 3.1, either $\bar{h}\gamma\bar{h}^{-1} = \gamma'$, or $\bar{h}\gamma\bar{h}^{-1} = \gamma'^{-1}$, then in the case under consideration $\bar{h}\gamma\bar{h}^{-1} = \gamma'$.

Let us consider case b):

$$\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}. \tag{14}$$

Note that $\bar{\varphi}'_0(S_g \times \{0\}) \overset{(9)}{=} \bar{h}(\bar{\varphi}(\bar{h}^{-1}(S_g \times \{0\}))) \overset{(14)}{=} \bar{h}(\bar{\varphi}(S_g \times \{0\})) \overset{(1)}{=} \bar{h}(S_g \times \{\frac{l}{k}\}) \overset{(14)}{=} S_g \times \{-\frac{l}{k}\}$. Taking into account the formula (11), we obtain that $S_g \times \{-\frac{l}{k}\} = S_g \times \{\frac{l'}{k} - j\}$. Since $-\frac{l}{k} \in (-0, 1]$ and $\frac{l'}{k} \in [0, 1)$, then the equality $\frac{l'}{k} - j = -\frac{l}{k}$ for

$l > 0$ is satisfied in the case of $j = 1$, that is,

$$\bar{\varphi}'_0 = \gamma'\bar{\varphi}'. \tag{15}$$

If $l = l' = 0$, then we get that $j = 0$, that is,

$$\bar{\varphi}'_0 = \bar{\varphi}'. \tag{16}$$

Therefore, either $l + l' = k$ or $l = l' = 0$.

Since $\bar{h}(\gamma(\bar{h}^{-1}(S_g \times \{0\}))) \overset{(14)}{=} \bar{h}(\gamma(S_g \times \{0\})) \overset{(2)}{=} \bar{h}(S_g \times \{-1\}) \overset{(14)}{=} S_g \times \{1\} \overset{(2)}{=} \gamma'^{-1}(S_g \times \{0\})$ and, according to Lemma 3.1, either $\bar{h}\gamma\bar{h}^{-1} = \gamma'$, or $\bar{h}\gamma\bar{h}^{-1} = \gamma'^{-1}$, then in the case under consideration $\bar{h}\gamma\bar{h}^{-1} = \gamma'^{-1}$.

□

Denote by

$$p_{S_g} \colon S_g \times \mathbb{R} \to S_g$$

the canonical projection defined by the formula $p_{S_g}(z, r) = z$.

We prove Theorem 1 which states that homeomorphisms φ and φ' are topologically conjugate if and only if $n = n'$, $k = k'$ and there exists a homeomorphism $H \colon S_g \to S_g$ such that one of the following conditions holds:

1. $l = l' \geq 0$, $HP = P'H$ and $HJ = J'H$;
2. $l = l' = 0$, $HP = P'H$ and $HJ = J'^{-1}H$;
3. $l + l' = k$, $HP = J'P'H$ and $HJ = J'^{-1}H$.

Proof. **Necessity.** Let homeomorphisms φ and φ' be topologically conjugate. Then, by Lemma 3.1, there exists the homeomorphism $h \colon M_J \to M_{J'}$ and its lift $\bar{h} \colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ such that $h\varphi = \varphi' h$ and $\bar{h}(S_g \times \{0\}) = S_g \times \{0\}$. By item 1 of Lemma 3.1, $k = k'$, $n = n'$. Therefore, $p_J^{-1}(NW(\varphi)) = p_{J'}^{-1}(NW(\varphi')) = S_g \times \mathcal{I}$. Since the homeomorphism h maps the connected components of the nonwandering set $NW(\varphi)$ to the connected components of the nonwandering set $NW(\varphi')$, for any $r \in \mathcal{I}$ there exists $r' \in \mathcal{I}$ such that $\bar{h}(S_g \times \{r\}) = S_g \times \{r'\}$. Then for any

$r \in \mathcal{I}$ the homeomorphism

$$H_r = p_{S_g}|_{\bar{h}(S_g \times \{r'\})} \bar{h} p_{S_g}^{-1}|_{S_g \times \{r\}} \colon S_g \to S_g$$

is well defined. That is, $\bar{h}(z, r) = (H_r, r')$, where $r, r' \in \mathcal{I}$.

We will prove that the homeomorphism H_r is isotopic to H_0 for any $r \in \mathcal{I}$.

We define a family of continuous mappings $F_{r,t} \colon S_g \to S_g$ by the formula $F_{r,t}(z) = p_{S_g}(\bar{h}(z, rt))$, where $t \in [0, 1]$, $r \in \mathcal{I}$. Then $F_{r,t}$ defines a homotopy connecting the mappings $F_{r,0} = H_0$ and $F_{r,1} = H_r$. Thus, the homeomorphisms H_0 and H_r are homotopic, and hence are isotopic [15, Sec. 5.15] for any $r \in \mathcal{I}$.

According to Lemma 3.2, three cases are possible. Let us consider each of them separately.

Consider item 1 of Lemma 3.2: $l = l' \geq 0$,

$$\bar{h}\gamma = \gamma' \bar{h}, \tag{17}$$

$$\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1} \tag{18}$$

and $\bar{h}(S_g \times \{r\}) = S_g \times \{r\}$ for any $r \in \mathcal{I}$.

Since $\gamma'^l(\bar{\varphi}'^k(S_g \times \{r\})) \overset{(13),(1)}{=} \gamma'^l(S_g \times \{r + l\}) \overset{(2)}{=} S_g \times \{r\}$ for any $r \in \mathcal{I}$ and $\gamma'^l \bar{\varphi}'^k \overset{(18)}{=} \gamma'^l \bar{h}\bar{\varphi}^k \bar{h}^{-1} \overset{(17)}{=} \bar{h}\gamma^l \bar{\varphi}^k \bar{h}^{-1}$, then $H_r J^l P^k H_r^{-1} = J'^l P'^k$ for any $r \in \mathcal{I}$. Since the homeomorphisms $J^l P^k$, $J'^l P'^k$ are pseudo-Anosov, H_r is isotopic to H_0 for any $r \in \mathcal{I}$, then from Proposition 2.4 it follows that $H_r = H_0$ for any $r \in \mathcal{I}$. Put $H = H_0$.

Since $\bar{h}(\bar{\varphi}(z, 0)) = (H(P(z)), \frac{l}{k})$, $\bar{\varphi}'(\bar{h}(z, 0)) = (P'(H(z)), \frac{l}{k})$ and $\bar{h}\bar{\varphi} = \bar{\varphi}'\bar{h}$, then $HP = P'H$.

Since $\bar{h}(\gamma(z, 0)) = (H(J(z)), -1)$, $\gamma'(\bar{h}(z, 0)) = (J'(H(z)), -1)$ and $\bar{h}\gamma = \gamma'\bar{h}$, then $HJ = J'H$.

Thus, it is proved that the condition 1 of the theorem is necessary for $l = l' \geq 0$ and $\bar{h}\gamma = \gamma'\bar{h}$.

Consider item 2 of Lemma 3.2: $l = l' = 0$,

$$\bar{h}\gamma = \gamma'^{-1}\bar{h},$$

$$\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1}$$

and $\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}$ for any $r \in \mathcal{I}$;

Since $\bar{h}(\bar{\varphi}(z,r)) = (H_r(P(z)), -r)$, $\bar{\varphi}'(\bar{h}(z,0)) = (P'(H_r(z)), -r)$ for any $r \in \mathcal{I}$ and $\bar{h}\bar{\varphi} = \bar{\varphi}'\bar{h}$, then $H_r P = P' H_r$ for any $r \in \mathcal{I}$. Since the homeomorphisms P, P' are pseudo-Anosov, H_r is isotopic to H_0 for any $r \in \mathcal{I}$, it follows from Proposition 2.4 that $H_r = H_0$ for any $r \in \mathcal{I}$. Put $H = H_0$.

Since $\bar{h}(\bar{\varphi}(z,0)) = (H(P(z)), \frac{l}{k})$, $\bar{\varphi}'(\bar{h}(z,0)) = (P'(H(z)), \frac{l}{k})$ and $\bar{h}\bar{\varphi} = \bar{\varphi}'\bar{h}$, then $HP = P'H$.

Since $\bar{h}(\gamma(z,0)) = (H(J(z)), 1)$, $\gamma'^{-1}(\bar{h}(z,0)) = (J'^{-1}(H(z)), 1)$ and $\bar{h}\gamma = \gamma'^{-1}\bar{h}$, then $HJ = J'^{-1}H$.

Thus, it is proved that the condition 2 of the theorem is necessary for $l = l' = 0$ and $\bar{h}\gamma = \gamma'^{-1}\bar{h}$.

Consider item 3 of Lemma 3.2: $l + l' = k$,

$$\bar{h}\gamma = \gamma'^{-1}\bar{h}, \tag{19}$$

$$\gamma'\bar{\varphi}' = \bar{h}\bar{\varphi}\bar{h}^{-1} \tag{20}$$

and $\bar{h}(S_g \times \{r\}) = S_g \times \{-r\}$ for any $r \in \mathcal{I}$.

Since $\gamma'^{k-l}(\bar{\varphi}'^k(S_g \times \{r\})) \overset{(1)}{=} \gamma'^{k-l}(S_g \times \{r+k-l\}) \overset{(2)}{=} S_g \times \{r\}$ for any $r \in \mathcal{I}$ and $\gamma'^{k-l}\bar{\varphi}'^k \overset{(20)}{=} \gamma'^{-l}\bar{h}\bar{\varphi}^k\bar{h}^{-1} \overset{(19)}{=} \bar{h}\gamma^l\bar{\varphi}^k\bar{h}^{-1}$, then $H_r J^l P^k H_r^{-1} = J'^{k-l}P'^k$ for any $r \in \mathcal{I}$. Since the homeomorphisms $J^l P^k$, $J'^{k-l}P'^k$ are pseudo-Anosov, H_r is isotopic to H_0 for any $r, \in \mathcal{I}$, then Proposition 2.4 implies that $H_r = H_0$ for any $r \in \mathcal{I}$. Put $H = H_0$.

Since $\bar{h}(\bar{\varphi}(z,0)) = (H(P(z)), \frac{l}{k})$, $\gamma'(\bar{\varphi}'(\bar{h}(z,0))) = (J'(P'(H(z))), \frac{l}{k})$ and $\bar{h}\bar{\varphi} = \gamma'\bar{\varphi}'\bar{h}$, then $HP = J'P'H$.

Since $\bar{h}(\gamma(z,0)) = (H(J(z)), -1)$, $\gamma'^{-1}(\bar{h}(z,0)) = (J'(H(z)), -1)$ and $\bar{h}\gamma = \gamma'^{-1}\bar{h}$, then $HJ = J'^{-1}H$.

Thus, it is proved that the condition 3 of the theorem is necessary for $l + l' = k$.

Sufficiency. Let $n = n'$, $k = k'$ and there exists a homeomorphism $H\colon S_g \to S_g$ such that $l = l' \geq 0$, $HP = P'H$ and $HJ = J'H$. We define a homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ by the formula $\bar{h}(z,r) = (H(z), r)$. Then it is directly verified that $\bar{h}\bar{\varphi} = \bar{h}\bar{\varphi}'$ and $\bar{h}\gamma = \gamma'\bar{h}$. Thus, according to Proposition 2.5, the

homeomorphism $\bar{h}$ projects into a homeomorphism $h\colon M_J \to M_{J'}$ and $h\varphi = \varphi' h$. The sufficiency of condition 1 of the theorem is proved.

Let us prove the sufficiency of conditions 2 and 3 of the theorem. We define a homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ by the formula $\bar{h}(z, r) = (H(z), -r)$. Similar to condition 1, under condition 2 (3) the equalities $\bar{h}\bar{\varphi} = \bar{h}\bar{\varphi}'$, $\bar{h}\gamma = \gamma'^{-1}\bar{h}$ ($\bar{h}\bar{\varphi} = \bar{h}\gamma'\bar{\varphi}'$, $\bar{h}\gamma = \gamma'^{-1}\bar{h}$) hold and the homeomorphism $\bar{h}$ projects to a homeomorphism $h\colon M_J \to M_{J'}$ such that $h\varphi = \varphi' h$.

$\square$

4 Mappings with Invariant One-dimensional Foliations

This section contains the proof of Theorem 2 and the auxiliary Lemma.

Let $f \in \mathcal{G}_*$. Note that from the ambient Ω-conjugacy of the homeomorphism f with a homeomorphism of class Φ (see Proposition 1.5) it follows that the connected components of the nonwandering set $NW(f)$ of the homeomorphism f are homeomorphic to a surface S_g of the same genus g and have the same period k. We put

$$f_0 = f^k.$$

Recall that $\mathcal{G}_* \subset \mathcal{G}$ is a subset of homeomorphisms satisfying the following condition. If $f \in \mathcal{G}_*$ is a homeomorphism on M^3, then on each connected component V of $M^3 \setminus NW(f)$ there exists a one-dimensional f_0-invariant foliation, each leaf of which is an open arc having exactly two boundary points lying on two different connected components of ∂V. The union of the closures of all such arcs on all connected components of $M^3 \setminus NW(f)$ forms an f-invariant one-dimensional foliation on M^3, which we denote by $\mathcal{L}_f$.

Lemma 4.1. *Let $f \in \mathcal{G}_*$ and V be a connected component of $M^3 \setminus (NW(f))$. Then on cl V there exists a two-dimensional f_0-invariant foliation $\mathcal{P}_V$, each leaf of which is homeomorphic to the surface S_g and intersects any leaf of $\mathcal{L}_f$ at exactly one point.*

Proof. Let $f \in \mathcal{G}_*$ be a homeomorphism with nonwandering set $NW(f)$ consisting of connected components homeomorphic to S_g and let V be a connected component of $M^3 \setminus NW(f)$. Denote by $\mathcal{L}_V$ the one-dimensional foliation on V, each leaf of which is a connected component of the intersection of a leaf of $\mathcal{L}_f$ with V.

By Proposition 1.5, the boundary of V consists of an attractor A and a repeller R of the homeomorphism f_0. Let l_z be the leaf of $\mathcal{L}_V$ that contains the point $z \in A$ in its closure and $w \in V \cap l_z$. Since the boundary of any leaf of the foliation $\mathcal{L}_V$ consists of exactly two points, one of which lies on the attractor A and the other on the repeller R, then a continuous mapping $\xi \colon V \to A$ is defined by the formula $\xi(w) = z$.

Thus, the triple (V, ξ, A) defines an orientable bundle of the total space V with projection ξ over the compact space A and a leaf homeomorphic to the interval $(0,1)$. It is known (see [16, Chapter 4, Theorem 4.3]) that an orientable bundle over a compact surface A with leaf homeomorphic to $\mathbb{R}$ at every point of A is isomorphic to the standard trivial bundle $(S_g \times \mathbb{R}, p_{S_g}, S_g)$, where

$$p_{S_g} \colon S_g \times \mathbb{R} \to S_g$$

is the canonical projection. Obviously, the standard trivial bundles $(S_g \times \mathbb{R}, p_{S_g}, S_g)$ and $(S_g \times (0,1), p_{S_g}, S_g)$ are isomorphic. Therefore, the bundle (V, ξ, A) is isomorphic to the standard trivial bundle $(S_g \times (0,1), p_{S_g}, S_g)$ and there exist homeomorphisms $h \colon V \to S_g \times (0,1)$ and $H \colon A \to S_g$ such that $p_{S_g} h = H\xi$. We define the foliation $\tilde{\mathcal{L}} = \{p_{S_g}^{-1}(z) \cap S_g \times [0,1] | z \in \ S_g\}$. Since the boundary of each leaf of the foliation $\mathcal{L}_V$ consists of two points lying on different connected components of the boundary of the set V, the homeomorphism h can be extended to the homeomorphism $h \colon cl\ V \to S_g \times [0,1]$ such that h maps the closure of each leaf $l_z \in \mathcal{L}_V$ to a leaf of the foliation $\tilde{\mathcal{L}}$.

We define a homeomorphism $\tilde{f} = h f_0 h^{-1} \colon S_g \times [0,1] \to S_g \times [0,1]$ and construct on the space $S_g \times [0,1]$ $\tilde{f}$- invariant foliation $\tilde{\mathcal{P}}$ such that each leaf of the foliation $\tilde{\mathcal{P}}$ is homeomorphic to S_g and intersects any leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point.

Without loss of generality, we can assume that $h(A) = S_g \times \{0\}$. Then the set $\tilde{A} = h(A)$ is an attractor of the homeomorphism $\tilde{f}$, $h(R) = S_g \times \{1\}$ and the set $\tilde{R} = h(R)$ is a repeller of the homeomorphism $\tilde{f}$. Since the foliation $\mathcal{L}_V$ is invariant under f_0 and h maps the closure of each leaf of the foliation $\mathcal{L}_V$ to a leaf of the foliation $\tilde{\mathcal{L}}$, the foliation $\tilde{\mathcal{L}}$ is invariant under $\tilde{f}$.

Let us prove that there exists a surface $\tilde{\Sigma} \subset S_g \times (0,1)$ of genus g that intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point and satisfies the condition $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$.

We put $\Sigma = S_g \times \left\{\frac{1}{2}\right\}$. If $\tilde{f}(\Sigma) \cap \Sigma = \emptyset$, then $\tilde{\Sigma} = \Sigma$. If $\tilde{f}(\Sigma) \cap \Sigma \neq \emptyset$, then we construct the surface $\tilde{\Sigma}$ as follows.

Let U be the closed subset of $S_g \times [0,1]$ bounded by the surfaces Σ and $\tilde{A}$. Since $\tilde{A}$ is an attractor of the homeomorphism $\tilde{f}$, there exists a number $r \in \mathbb{N}$

such that $\tilde{f}^{2^r}(U) \subset int\ U$. We prove the existence of the surface $\tilde{\Sigma}$ by induction on $r \in \mathbb{N}$.

1) Let $r = 1$. Then

$$\tilde{f}^2(\Sigma) \cap \Sigma = \emptyset \text{ and } \tilde{f}^2(\Sigma) \subset int\ U. \tag{21}$$

Let $U' = U \cup \tilde{f}(U)$, $\bar{\Sigma} = \partial(U')$ (in Fig. 1 the surface $\bar{\Sigma}$ is shown by a solid line), $\Sigma' = \tilde{f}(\Sigma)$, $\Lambda = \Sigma \cap \bar{\Sigma}$ and $\Lambda' = \Sigma' \cap \bar{\Sigma}$ (see Fig. 1). Then $\bar{\Sigma} = \Lambda \cup \Lambda'$.

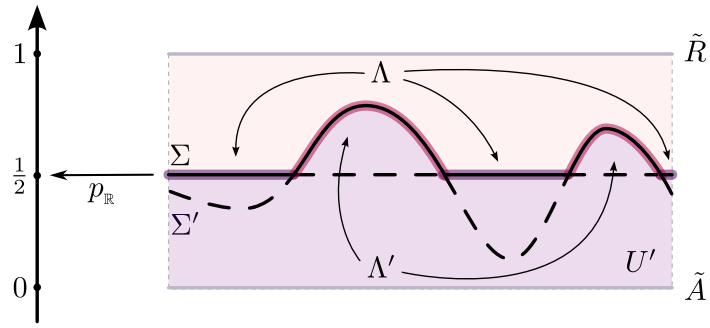

Figure 1: Surfaces Σ и Σ'.

We prove that the surface $\bar{\Sigma}$ intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point. Assume the contrary. Let there exist a leaf $\tilde{l}$ of the foliation $\tilde{\mathcal{L}}$ intersecting the surface $\bar{\Sigma}$ at more than one point. Since the foliation $\tilde{\mathcal{L}}$ is invariant under $\tilde{f}$ and the surface Σ intersects each leaf of the foliation $\tilde{\mathcal{L}}$ transversally and at a unique point, the surface Σ' also intersects each leaf of the foliation $\tilde{\mathcal{L}}$ transversally and at a unique point. Therefore, the intersection $\tilde{l} \cap \bar{\Sigma}$ consists of two points $\tilde{w} = \tilde{l} \cap \Lambda$ and $\tilde{w}' = \tilde{l} \cap \Lambda'$. Then the leaf $\tilde{l}$ is divided into 3 curves. The first curve has boundary points $\tilde{l} \cap \tilde{A}$, $\tilde{w}$ and is contained in the domain U'. The second curve has boundary points $\tilde{w}$, $\tilde{w}'$ and lies outside the domain U'. Since the fiber $\tilde{l}$ intersects the surface $\bar{\Sigma}$ transversally and, by assumption, at exactly two distinct points, the third curve with boundary points $\tilde{w}'$ and $\tilde{l} \cap \tilde{R}$ is contained in the domain U'. We obtain a contradiction with the fact that $U' \cap \tilde{R} = \emptyset$.

We will show that the surface $\bar{\Sigma}$ is transformed by continuous perturbation along the leaves into a surface $\tilde{\Sigma}$ of genus g such that $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$.

We set $U'' = \tilde{f}(U) \cup \tilde{f}^2(U)$, $\bar{\Sigma}' = \partial U''$ (In Fig. 2 the surface $\bar{\Sigma}'$ is depicted by

a solid line). As for the surface $\bar{\Sigma}$, it can be proven that the surface $\bar{\Sigma}'$ intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point.

Let

$$p_{\mathbb{R}} : S_g \times \mathbb{R} \to \mathbb{R}$$

denote the canonical projection. For each leaf $\tilde{l}$ intersecting the set Λ and the surface $\bar{\Sigma}'$ at points $\tilde{w}$ and $\tilde{w}'$, respectively, let $\tilde{w}_*$ denote the point on the leaf $\tilde{l}$ such that $p_{\mathbb{R}}(\tilde{w}_*) = \frac{1}{2}(p_{\mathbb{R}}(\tilde{w}) + p_{\mathbb{R}}(\tilde{w}'))$. The point $\tilde{w}_*$ is the midpoint of the segment lying on the leaf $\tilde{l}$ with endpoints $\tilde{w}$ and $\tilde{w}'$. Let Λ_* denote the set of all midpoints of segments lying on the leaves of the foliation $\tilde{\mathcal{L}}$ with endpoints on the surfaces Λ and $\bar{\Sigma}'$ (see Fig. 2).

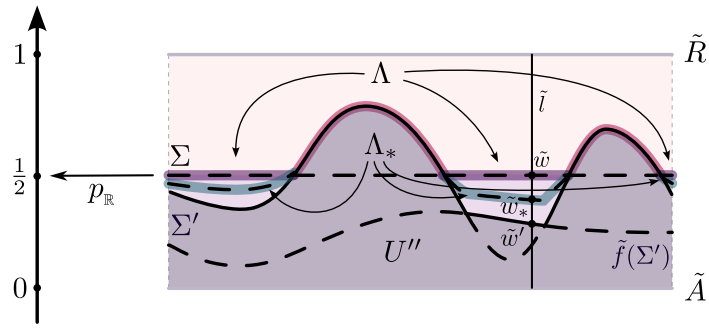

Figure 2: The set Λ_*.

We put $\tilde{\Sigma} = \Lambda_* \cup \Lambda'$. By construction, the surface $\tilde{\Sigma}$ is homeomorphic to the surface Σ' of genus g and intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point.

Denote by $\tilde{U}$ the subspace of $S_g \times [0,1]$ bounded by the surfaces $\tilde{A}$ and $\tilde{\Sigma}$. If $\tilde{w} \in \Lambda'$, then $\tilde{f}(\tilde{w}) \in \Sigma'' \subset int\ \tilde{U}$. If $\tilde{w} \in \Lambda_*$, then $\tilde{w} \in int\ U$ and $\tilde{f}(\tilde{w}) \in int\ \tilde{f}(U) \subset int\ \tilde{U}$. Thus, $\tilde{f}(\tilde{\Sigma}) \subset int\ \tilde{U}$ and $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$. The induction base is proved.

2) Suppose that for $r = k$ there exists a surface $\tilde{\Sigma}$ of genus g that intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point and $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$.

3) Let $r = k+1$. Put $\tilde{g} = \tilde{f}^{2^k}$. Then $\tilde{g}^2(\Sigma) \cap \Sigma = \emptyset$ and, by the induction base ($r = 1$), there exists a surface $\tilde{\Sigma}' \subset S_g \times (0,1)$ of genus g such that $\tilde{\Sigma}'$ intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point and $\tilde{g}(\tilde{\Sigma}') \cap \tilde{\Sigma}' = \emptyset$. Consequently, $\tilde{f}^{2^k}(\tilde{\Sigma}') \cap \tilde{\Sigma}' = \emptyset$. It follows from the induction hypothesis that there exists a

surface $\tilde{\Sigma} \subset S_g \times (0,1)$ of genus g such that $\tilde{\Sigma}$ intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point and $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$. The statement of induction is proven.

Let us construct the foliation $\tilde{\mathcal{P}}$. Let $\tilde{\Sigma} \subset S_g \times (0,1)$ be a surface of genus g that intersects each leaf of the foliation $\tilde{\mathcal{L}}$ at a unique point and satisfies the condition $\tilde{f}(\tilde{\Sigma}) \cap \tilde{\Sigma} = \emptyset$, $\tilde{U}$ be a closed subset of $S_g \times [0,1]$ bounded by the surfaces $\tilde{A}$ and $\tilde{\Sigma}$. Then $\tilde{f}(\tilde{\Sigma}) \subset int\ \tilde{U}$ and the set $K = \tilde{U} \setminus \tilde{f}(\tilde{U})$ is the fundamental domain of $\tilde{f}$ on $S_g \times (0,1)$, that is, $\bigcup\limits_{i \in \mathbb{Z}} \tilde{f}^i(K) = S_g \times (0,1)$ and $\tilde{f}^i(K) \cap \tilde{f}^j(K) = \emptyset$ for $i \neq j$. For the leaf $\tilde{l} \in \tilde{\mathcal{L}}$ we put $\tilde{l}_K = \tilde{l} \cap cl\ K$, $\tilde{w}_0 = \tilde{l} \cap \tilde{f}(\tilde{\Sigma})$ and $\tilde{w}_1 = \tilde{l} \cap \tilde{\Sigma}$. We parametrize the curve $\tilde{l}_K$ by assigning to the point $\tilde{w} \in \tilde{l}_K$ the parameter $t = \frac{p_{\mathbb{R}}(\tilde{w}) - p_{\mathbb{R}}(\tilde{w}_0)}{p_{\mathbb{R}}(\tilde{w}_1) - p_{\mathbb{R}}(\tilde{w}_0)}$, equal to the ratio of the length of the segment between the points $\tilde{w}_0$ and $\tilde{w}$ to the length of the segment $\tilde{l}_K$. We denote this parametrization by $\rho\colon cl\ K \to [0,1]$. Then $\{\tilde{A}, \tilde{R}, \tilde{f}^i(\rho^{-1}(t)) | i \in \mathbb{Z}, t \in [0,1]\}$ is the desired foliation of $\tilde{\mathcal{P}}$.

Since the homeomorphisms $f_0|_{cl\ V}$ and $\tilde{f}$ are topologically conjugate by the homeomorphism h, then $\mathcal{P}_V = h^{-1}(\tilde{\mathcal{P}})$ is a two-dimensional f_0-invariant foliation, each leaf of which is homeomorphic to the surface S_g and intersects the leaves of the foliation $\mathcal{L}_f$ at exactly one point.

□

Let us prove Theorem 2 which states that a homeomorphism $f \in \mathcal{G}$ is topologically conjugate to a homeomorphism from the class Φ if and only if $f \in \mathcal{G}_*$.

Proof. Since the homeomorphism $\varphi \in \Phi$ has a one-dimensional invariant foliation $\mathcal{L}_\varphi = \{p_J(z \times \mathbb{R}) | z \in S_g\}$, then $\Phi \subset \mathcal{G}_*$.

We will prove that the homeomorphism $f \in \mathcal{G}_*$ is topologically conjugate to some homeomorphism from the class Φ.

Let f be a homeomorphism from the class $\mathcal{G}_*$. According to Proposition 1.5, without loss of generality we can assume that f is defined on the manifold M_J with the natural projection p_J and coincides with some homeomorphism $\varphi = \varphi_{P,J,n,k,l}\colon M_J \to M_J$ from the class Φ on their common nonwandering set. Moreover, its lift $\bar{f}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ coincides with the homeomorphism $\bar{\varphi} = \bar{\varphi}_{P,J,n,k,l}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ on the set $p_J^{-1}(NW(f)) = S_g \times \mathcal{I}$, where $\mathcal{I} = \{\frac{i}{2nk}, i \in \mathbb{Z}\}$.

Let us construct a homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ such that $\bar{h}\gamma = \gamma\bar{h}$ and $\bar{h}\bar{f} = \bar{\varphi}_{P,J,n,k,l}\bar{h}$.

First let us consider the case $k = 1$, i.e. the homeomorphism $\varphi_{P,n,1,0}$.

It follows from Lemma 4.1 that the homeomorphism f has an f-invariant two-dimensional foliation $\mathcal{P}_f$, each leaf of which is homeomorphic to S_g and intersects the connected components of the foliation $\mathcal{L}_f \cap (M_J \setminus NW(f))$ at a unique point. Then the lifts $\mathcal{P}_{\bar{f}}$ and $\mathcal{L}_{\bar{f}}$ of the foliations $\mathcal{P}_f$ and $\mathcal{L}_f$ to $S_g \times \mathbb{R}$ are invariant under the homeomorphism $\bar{f}$ and the leaves of these foliations intersect at a unique point. Note that the connected components of the set $S_g \times \mathcal{I}$ are leaves of the foliation $\mathcal{P}_{\bar{f}}$.

For $i \in \{0, 1, \dots, 2n-1\}$, we set $\bar{V}_i = S_g \times (\frac{i}{2n}, \frac{i+1}{2n})$ and $\bar{\mathcal{B}}_i = S_g \times \{\frac{i}{2n}\}$. We construct a homeomorphism $H_i \colon cl\ \bar{V}_i \to cl\ \bar{V}_i$, $i \in \{0, 1, \dots, 2n-1\}$, conjugating the mappings $\bar{f}|_{cl\ \bar{V}_i}$ and $\bar{\varphi}|_{cl\ \bar{V}_i}$. We denote by $\mathcal{L}_i$, $\mathcal{P}_i$ the restriction of the foliations $\mathcal{L}_{\bar{f}}$, $\mathcal{P}_{\bar{f}}$ to $cl\ \bar{V}_i$. Let $\tilde{\mathcal{L}}_i = \{z \times [\frac{i}{2n}, \frac{i+1}{2n}]\}_{z \in S_g}$, $\tilde{\mathcal{P}}_i = \{S_g \times \{r\}\}_{r \in [\frac{i}{2n}, \frac{i+1}{2n}]}$. The foliations $\tilde{\mathcal{L}}_i$ and $\tilde{\mathcal{P}}_i$ are invariant under the homeomorphism $\bar{\varphi}$.

For a point $z \in \bar{\mathcal{B}}_i$, let l_z be the leaf of the foliation $\mathcal{L}_i$ such that $z \in cl\ l_z$. Fix a point $z \in \bar{\mathcal{B}}_i$. Let Σ_x be the leaf of the foliation $\mathcal{P}_i$ passing through the point $x \in cl\ l_z$. Fix a point $x \in l_z$. Put $y = l_z \cap \bar{f}(\Sigma_x)$. Denote by $\sigma \subset l_z$ the closed arc bounded by the points x and y. Introduce similar notations with a wavy line for the homeomorphism $\bar{\varphi}$, choosing $\tilde{z} = z$. Let $\mu_\sigma \colon \sigma \to \tilde{\sigma}$ be a homeomorphism such that $\mu_\sigma(x) = \tilde{x}$ (see Fig. 3).

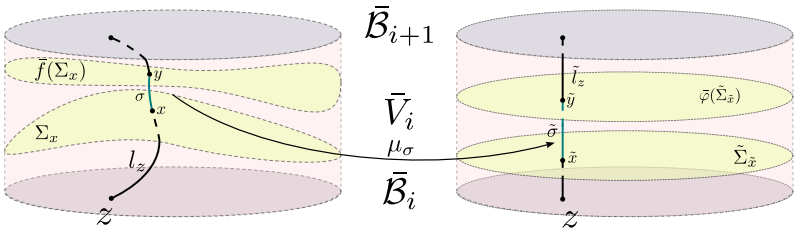

Figure 3: The homeomorphism μ_σ.

Since the topological space bounded by the leaves Σ_x, $\bar{f}(\Sigma_x) = \Sigma_y$ and containing the arc σ, is a fundamental domain of the action of f on $\bar{V}_i$, then for any point $w \in l_z$ there exists a unique integer m such that $(\bar{f})^m(\Sigma_w) \cap (\sigma \setminus \{y\}) \neq \emptyset$. We define the homeomorphism $\mu \colon l_z \to \tilde{l}_z$ by the formula

$$\mu(w) = \bar{\varphi}^{-m}(\tilde{\Sigma}_{\mu_\sigma(\bar{f}^m(\Sigma_w) \cap \sigma)}) \cap \tilde{l}_z$$

and, by continuity, extend the constructed mapping to the homeomorphism $\mu\colon cl\ l_z \to cl\ \tilde{l}_z$ (see Fig. 4).

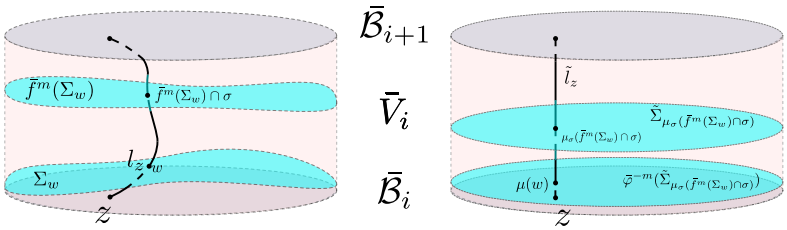

Figure 4: The homeomorphism μ.

Note that for any point $a \in cl\ \bar{V}_i$ there exists a unique pair of points $z_a \in \bar{\mathcal{B}}_i$, $w_a \in cl\ l_z$ such that $a = l_{z_a} \cap \Sigma_{w_a}$. We define a homeomorphism $\bar{h}_i\colon cl\ \bar{V}_i \to cl\ \bar{V}_i$ by the formula

$$\bar{h}_i(a) = \tilde{l}_{z_a} \cap \tilde{\Sigma}_{\mu(w_a)}$$

(see Fig. 5). It is directly verified that $\bar{h}_i \bar{f}|_{cl\ \bar{V}_i} = \bar{\varphi}|_{cl\ \bar{V}_i} \bar{h}_i$.

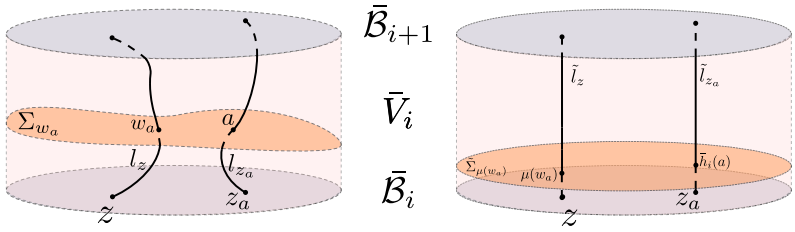

Figure 5: The homeomorphism $\bar{h}_i$.

We define homeomorphisms $H_i, H_{i+1}\colon S_g \to S_g$ by the formulas

$$H_i = p_{S_g}|_{\bar{h}(S_g\times\{\frac{i}{2n}\})}\bar{h}_i p_{S_g}^{-1}|_{S_g\times\{\frac{i}{2n}\}},$$

$$H_{i+1} = p_{S_g}|_{\bar{h}(S_g\times\{\frac{i+1}{2n}\})}\bar{h}_i p_{S_g}^{-1}|_{S_g\times\{\frac{i+1}{2n}\}},$$

where $p_{S_g} : S_g \times \mathbb{R} \to S_g$ is the natural projection. That is, $\bar{h}_i(z, \frac{i}{2n}) = (H_i(z), \frac{i}{2n})$ and $\bar{h}_i(z, \frac{i+1}{2n}) = (H_{i+1}(z), \frac{i+1}{2n})$.

By construction, $H_i = id_{S_g}$. Let us show that $H_{i+1} = id_{S_g}$.

First, let us prove that the homeomorphism H_i is homotopic to H_{i+1}. We define

a family of continuous mappings $F_t\colon S_g \to S_g$ ($t \in [0,1]$) by the formula

$$F_t(z) = p_{S_g}\Big(\bar{h}_i\Big(z, \frac{1}{2nk}(i+t)\Big)\Big).$$

Then the family F_t defines a homotopy connecting the mappings $F_0 = H_i$ and $F_1 = H_{i+1}$. Thus, the homeomorphisms H_i and H_{i+1} are homotopic.

Since $\bar{h}_i\Big(\bar{\varphi}\Big(z, \frac{i+1}{2n}\Big)\Big) = \bar{h}_i\Big(P(z), \frac{i+1}{2n}\Big) = \Big(H_{i+1}\Big(P(z)\Big), \frac{i+1}{2n}\Big)$, $\bar{f}\Big(\bar{h}_i\Big(z, \frac{i+1}{2n}\Big)\Big) = \bar{f}\Big(H_{i+1}(z), \frac{i+1}{2n}\Big) = \Big(P\Big(H_{i+1}(z)\Big), \frac{i+1}{2n}\Big)$ and, by construction, $\bar{h}_i\bar{\varphi}|_{cl\ \bar{V}_i} = \bar{f}|_{cl\ \bar{V}_i}\bar{h}_i$, then $H_{i+1}P = PH_{i+1}$. Since the homeomorphism H_{i+1} is homotopic to the identity mapping id_{S_g} and the homeomorphism P is a pseudo-Anosov mapping, then, according to Proposition 2.4, $H_{i+1} = id_{S_g}$.

Let $\bar{H}\colon S_g \times [0,1] \to S_g \times [0,1]$ be the homeomorphism coinciding with $\bar{h}_i$ on $cl\ \bar{V}_i$ for any $i \in \{0, 1, \ldots, 2n-1\}$ and let $[r]$ be the integer part of $r \in \mathbb{R}$. We define the homeomorphism $\bar{h}$ by the formula

$$\bar{h}(z,r) = \gamma^{-[r]}\bar{H}\gamma^{[r]}(z,r).$$

Since $\bar{h}\gamma = \gamma\bar{h}$, the homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ projects to the homeomorphism $h\colon M_J \to M_J$ (see Proposition 2.5). By construction, the homeomorphism $\bar{H}$ conjugates the homeomorphisms $\bar{f}$ and $\bar{\varphi}$ on $S_g \times [0,1]$. It is directly verified that $\bar{h}\bar{f} = \bar{\varphi}\bar{h}$. Consequently, $hf = \varphi h$.

Consider the homeomorphism $\varphi = \varphi_{P,J,n,k,l}$ for $k > 1$. Homeomorphisms

$$\gamma^l\bar{f}^k, \gamma^l\bar{\varphi}^k\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$$

are lifts of homeomorphisms

$$f^k, \varphi^k\colon M_J \to M_J$$

and $\gamma^l(\bar{f}^k(z,r)) = \gamma^l(\bar{\varphi}^k(z,r)) = (J^l(P^k(z)), r)$ for any $r \in \mathcal{I}$, where $J^lP^k\colon S_g \to S_g$ is a pseudo-Anosov homeomorphism. By the previous point ($k = 1$) there exists a homeomorphism $\tilde{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ such that $\tilde{h}\gamma = \gamma\tilde{h}$ and $\tilde{h}\gamma^l\bar{f}^k = \gamma^l\bar{\varphi}^k\tilde{h}$. It follows immediately that $\tilde{h}\bar{f}^k = \bar{\varphi}^k\tilde{h}$.

For $j = 0, \ldots, k-1$ we define the sets $\bar{W}_j = S_g \times [\frac{jl}{k}, \frac{jl}{k} + \frac{1}{k}]$ and the homeomor-

phism $\bar{h}_j\colon \bar{W}_j \to \bar{W}_j$ by the formula $\bar{h}_j = \bar{\varphi}^j \tilde{h}(\bar{f})^{-j}$. We set $\bar{W} = \bar{W}_0 \cup \cdots \cup \bar{W}_{k-1}$ and denote by $\bar{H}\colon \bar{W} \to \bar{W}$ the homeomorphism coinciding with $\bar{h}_j$ on $\bar{W}_j$ for $j = 0, \ldots, k-1$.

For $r \in \mathbb{R}$, let $m(r)$ be an integer such that $(z, r - m(r)) \in \bar{W}$. We define a homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ by the formula

$$\bar{h}(z, r) = \gamma^{-m(r)} \bar{H} \gamma^{m(r)}(z, r).$$

It is directly verified that $\bar{h}\bar{f} = \bar{\varphi}\bar{h}$. Since $\bar{h}\gamma = \gamma\bar{h}$, the homeomorphism $\bar{h}\colon S_g \times \mathbb{R} \to S_g \times \mathbb{R}$ projects onto the homeomorphism $h\colon M_J \to M_J$ (see Proposition 2.5). Therefore, $hf = \varphi h$.

$\square$

5 Generalized Pseudo-Anosov Dynamics

Denote by $P_0\colon \mathbb{S}^2 \to \mathbb{S}^2$ the orientation-preserving generalized pseudo-Anosov homeomorphism of the two-dimensional sphere with dilatation λ (the existence of such a homeomorphism follows from Proposition 2.2). Then the homeomorphism $P_1 = P_0^2$ is an orientation-preserving generalized pseudo-Anosov homeomorphism of the two-dimensional sphere with dilatation λ^2.

Since the dilatation of generalized pseudo-Anosov homeomorphisms is a topological invariant and there exist exactly two isotopy classes of homeomorphisms on the sphere $\mathbb{S}^2$ (one of which consists of all orientation-preserving maps, and the other of all orientation-reversing maps), then $P_0, P_1\colon \mathbb{S}^2 \to \mathbb{S}^2$ are topologically disjoint but isotopic generalized pseudo-Anosov homeomorphisms. Consequently, there exists an isotopy $F_t\colon \mathbb{S}^2 \to \mathbb{S}^2$ ($t \in [0,1]$) such that $F_0 = P_0$ and $F_1 = P_1$.

Let us associate each point $s \in \mathbb{S}^1$ with the parameter $r \in [0,1)$ such that $r \in p^{-1}(s(r))$, where $p\colon \mathbb{R} \to \mathbb{S}^1$ is the covering and $s(r) \in \mathbb{S}^1$ is the point corresponding to the parameter r. We define a homeomorphism $f\colon \mathbb{S}^2 \times \mathbb{S}^1 \to \mathbb{S}^2 \times \mathbb{S}^1$ by the formula

$$f(z, s(r)) = \begin{cases} (F_{2r}(z), \varphi_{1,1,0}(s(r))), & \text{if } r \in \left[0, \dfrac{1}{2}\right) \\ (F_{-2r+2}(z), \varphi_{1,1,0}(s(r))), & \text{if } r \in \left[\dfrac{1}{2}, 1\right) \end{cases}.$$

Since the diffeomorphism $\varphi_{1,1,0}\colon \mathbb{S}^1 \to \mathbb{S}^1$ is a source-sink mapping with source at $p(0)$ and sink at $p(\frac{1}{2})$, it is directly verified that the nonwandering set of the constructed mapping f consists of two cylindrically embedded spheres $\mathbb{S}^2_R = \mathbb{S}^2 \times \{p(0)\}$ and $\mathbb{S}^2_A = \mathbb{S}^2 \times \left\{p\left(\frac{1}{2}\right)\right\}$ such that the sphere $\mathbb{S}^2_R$ is a repeller for f and the sphere $\mathbb{S}^2_A$ is an attractor for f. The mapping $f|_{\mathbb{S}^2_R}\colon \mathbb{S}^2_R \to \mathbb{S}^2_R$ is topologically conjugate to P_0 and the mapping $f|_{\mathbb{S}^2_A}\colon \mathbb{S}^2_A \to \mathbb{S}^2_A$ is topologically conjugate to P_1. Therefore, homeomorphisms $f|_{\mathbb{S}^2_R}$ and $f|_{\mathbb{S}^2_A}$ are not topologically conjugate.

The constructed example shows that homeomorphisms of three-dimensional manifolds with generalized pseudo-Anosov mappings in the nonwandering set have more complicated dynamics, different from the dynamics of mappings of class $\mathcal{G}$ with pseudo-Anosov homeomorphisms in the nonwandering set.

Conclusion

This work completely describes the dynamics of orientation-preserving homeomorphisms on three-dimensional manifolds with a nonwandering set consisting of a finite number of cylindrically embedded two-dimensional surfaces, the restriction of some power of mapping to each of which is an orientation-preserving pseudo-Anosov homeomorphism. It is proved that the dynamics of homeomorphisms of the class under study is similar to the dynamics of model mappings. The main result of the thesis is the finding of necessary and sufficient conditions for the topological conjugacy of such homeomorphisms with model mappings. It is shown that the obtained results do not generalize to homeomorphisms with generalized pseudo-Anosov homeomorphisms in a nonwandering set.

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